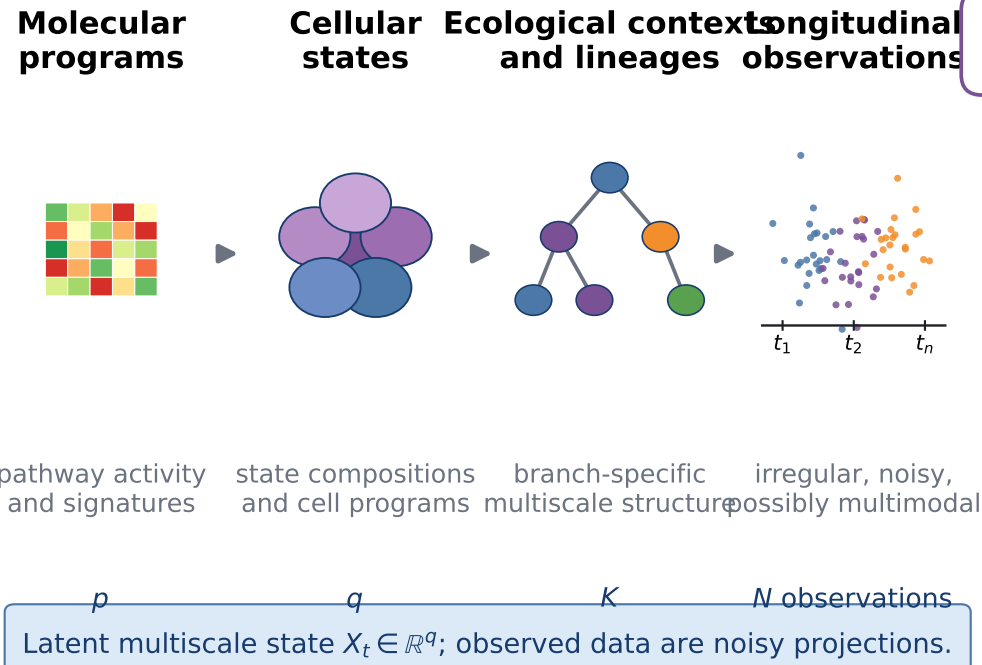


Figure 1. Unified Ornstein-Uhlenbeck-Lévy-Branching architecture

A general stochastic framework for multiscale cancer evolution under therapy, jumps, branching, and observation error

A Multiscale state representation



B Unified latent process

$$dX_t = \Theta_{B_t} \{\mu_{B_t}(t) - X_t\} dt + \Sigma_{B_t} dW_t + dJ_t$$

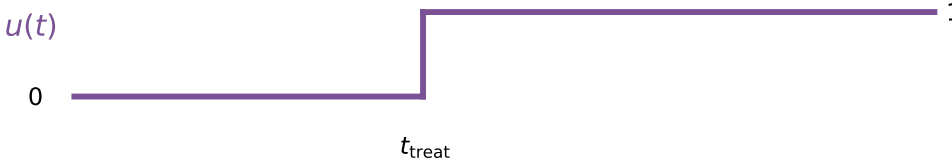
- Θ_{B_t} branch-specific restoring matrix
- $\mu_{B_t}(t)$ possibly time-varying attractor
- Σ_{B_t} branch-specific diffusion scale
- W_t Wiener process
- J_t compound-Poisson / Lévy-like jump process

Deterministic reversion + continuous diffusion + discontinuous jumps, all conditional on the current branch state.

C Therapy-dependent attractor shift

$$\mu_{B_t}(t) = \mu_{0, B_t} + \Delta_{B_t} u(t)$$

- μ_{0, B_t} pre-therapy branch attractor
- Δ_{B_t} therapy-induced displacement
- $u(t)$ treatment exposure or dose



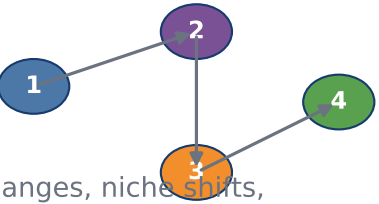
Step functions, continuous dose functions, and branch-specific responses are all allowed.

D Branch-state (lineage) process

$$B_t \in \{1, \dots, K\}, \quad B_t \sim \text{CTMC}(Q)$$

$$q_{kl} \geq 0 \ (k \neq l)$$

$$q_{kk} = -\sum_{\ell \neq k} q_{k\ell}$$



Branch transitions represent lineage changes, niche shifts, state conversion, or ecological reorganization.

E Observation model

$$Y_i = H_i X_{t_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, R_i)$$

- Y_i observed data at time, site, or platform
- H_i projection / loading / cross-scale mapping
- R_i measurement-error covariance

Supports irregular sampling, heterogeneous modalities, missing observations, and known measurement variance.

F Nested special cases

0	Brownian motion	$\Theta = 0, J_t = 0$	diffusion only
1	Standard OU	$\Theta > 0, J_t = 0$	mean reversion
2	Treatment-shifted OU	$\mu(t) = \mu_0 + \Delta u(t)$	therapy response
3	OU with jumps	$J_t \neq 0$	rare discontinuities
4	OU with branching	B_t varies	state switching
5	Full OULB	$\Theta_{B_t}, \mu_{B_t}(t), \Sigma_{B_t}, J_t$	all mechanisms

More general
Simpler

The full model is a coherent superset of all simpler cases.

Key takeaway

OULB unifies constrained stochastic evolution, therapy-driven attractor shifts, rare jumps, lineage branching, cross-scale mapping, and noisy observation within one interpretable generative framework.